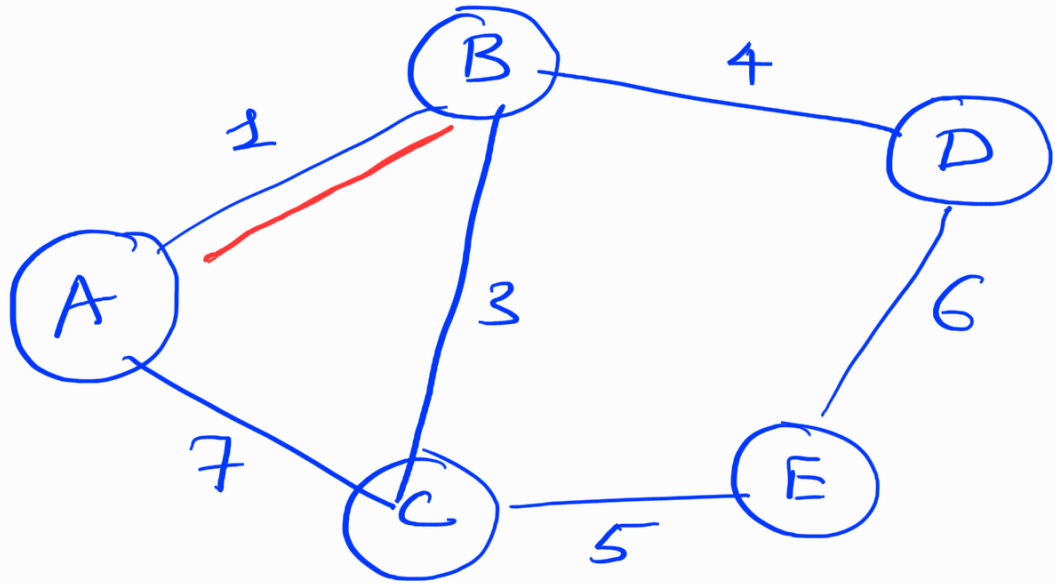


"Given a weighted graph  $G(V, E)$ ,  
Find out shortest path from starting  
vertex  $s$  to all other vertices"



$$S = \{A\}$$

Distance set

$$\checkmark d(A, B) = 1, \quad d(A, C) = 7, \quad d(A, D) = \infty$$

$$d(A, E) = \infty$$

Similarity with prim's :

- ① Maintain a set of covered vertices
- ② Maintain a set of visible edges
- ③ Calculate distance -

## Iteration 2

$$S = \{A, B\}$$

$$\text{visible edges} = \{AC, BC, BD\}$$

$$d(A, B) = 1, \quad d(A, C) \xrightarrow{\Delta \text{ min}} \begin{cases} d(A, C) = 7 \\ d(A, B) + d(B, C) \end{cases}$$

$$d(A, C) = 4, \quad d(A, D) = \overset{1}{d(A, B)} + \overset{3}{d(B, D)} = 4$$

$$= 1 + 4$$

$$= 5$$

$$d(A, E) = \infty$$

## Dijkstra's Algorithm

**Input:** Graph  $G = (V, E)$ , directed or undirected; positive edge lengths  $\{l_e; e \in E\}$ ; vertex  $s \in V$

**Output:** For all vertices  $u$  reachable from  $s$ ,  $\text{dist}(u)$  is set to the distance from  $s$  to  $u$ .

**procedure** DIJKSTRA( $G, l, s$ )

**for all**  $u \in V$  **do**

$\text{dist}(u) = \infty$

$\text{prev}(u) = \text{nil}$

$\text{dist}(s) = 0$

$H = \text{MAKEQUEUE}(V)$

$\triangleright$  using dist-values as keys

**while**  $H$  is not empty **do**

$u = \text{DELETETMIN}(H)$

**for all** edges  $(u, v) \in E$  **do**

**if**  $\text{dist}(v) > \text{dist}(u) + l(u, v)$  **then**

$\text{dist}(v) = \text{dist}(u) + l(u, v)$

$\text{prev}(v) = u$

$\text{DECREASEKEY}(H, v)$

# Prim's Algorithm

Procedure prims

$G$  – input graph

$U$  – random vertex

$V$  – vertices in graph  $G$

begin

$T = \emptyset$ ;

$U = \{1\}$ ;

    while ( $U \neq V$ )

        let  $(u, v)$  be the least cost edge such that  $u \in U$  and  $v \in V - U$ ;

$T = T \cup \{(u, v)\}$

$U = U \cup \{v\}$

end procedure

# Kruskal's Algorithm

- MAKE-SET( $v$ ) puts  $v$  in a set by itself
- FIND-SET( $v$ ) returns the name of  $v$ 's set
- UNION( $u, v$ ) combines the sets that  $u$  and  $v$  are in

MST-Kruskal( $G, w$ )

- 1  $A \leftarrow \emptyset$
- 2 for each vertex  $v \in V[G]$
- 3     do MAKE-SET( $v$ )
- 4 sort the edges of  $E$  into nondecreasing order by weight  $w$
- 5 for each edge  $(u, v) \in E$ , taken in nondecreasing order by weight
- 6     do if FIND-SET( $u$ )  $\neq$  FIND-SET( $v$ )
- 7         then  $A \leftarrow A \cup \{(u, v)\}$
- 8         UNION( $u, v$ )
- 9 return  $A$

